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Electrochemical characteristics of heterogeneous bipolar membranes and electromembrane process of recovery of nitric acid and sodium hydroxide from sodium nitrate solution

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ABSTRACT

Frequency spectra of electrochemical impedance and current-overvoltage characteristics of an MB-2m bipolar membrane containing ion-polymer with catalytically active phosphoric acid groups in water dissociation reaction, and industrial MB-1, MB-2, and MB-3 bipolar membranes were studied in a system containing nitric acid and sodium hydroxide solutions. It is shown that the electrical resistance and overvoltage of bipolar region in the modified MB-2m and industrial MB-3 membranes are close and significantly lower than in industrial MB-1 and MB-2membranes. The effective rate constants of catalytic reaction of water dissociation are determined in the studied membranes. The process of producing nitric acid and sodium hydroxide from sodium nitrate with the modified MB-2m bipolar membrane and industrial MB-1 and MB-3 bipolar membranes was studied. It is shown that the use of the MB-2m bipolar membrane allows producing nitric acid and sodium hydroxide with higher concentrations, current efficiencies, specific productivity, lower specific energy consumption, and degree of sodium nitrate contamination than in the case when using industrial MB-1 and MB-3 membranes.

1. Introduction

Along with the processes of desalination, deionization, concentration, and separation of electrolytes, electrodialysis with bipolar membranes (BMED) has found wide practical application [1–10]. BMED is currently used to produce acids, alkalis, and bases from solutions of inorganic and organic salts [11–23], pH correction of natural waters, technological solutions in the chemical and food industries [5–7,17,18,24], for producing deionized and ultrapure water [25–28], for hydrogen producing based on ethanol-type fermentation and bipolar membrane electrodialysis [29], in the purification of gases from acidic impurities [30,31], and in electro-membrane synthesis [32–34]. In addition, bipolar electrodialysis allows removing boron from wastewater [35,36], and regenerating adsorbent granules saturated with hydrogen sulfide [37].

Industrial bipolar membranes (BPM) are divided into homogeneous and heterogeneous. The homogeneous BPMs have higher electrochemical characteristics: lower operating voltage and higher current output of hydrogen and hydroxyl ions. A review of methods for their preparation and characteristics are given in [3,6]. The heterogeneous bipolar membranes are produced by hot pressing or by rolling the cation exchange and anion exchange membranes [38–40] as well as by applying an ion-polymer with oppositely charged ionogenic groups on industrial monopolar membranes [41–43].

Various catalytic additives are introduced into the bipolar region of heterogeneous bipolar membranes to reduce membrane voltage. Ion-polymers [9,41,44–48] or hydroxides and salts of heavy metals [49–57], and metal-organic frameworks (MOFs) [58] are used as catalysts. A powerful electric field with a strength of 10^7–10^8 V/m [59,60] arises at the internal interphase boundary (in the bipolar region) [61] of the BPM when an electric current flows through it. An electrocatalytic reaction of water dissociation with a rate many degrees higher than the rate of water dissociation in water and water solutions when these additives are presented runs in these conditions [59,60,62,63]. This leads to a decrease in overvoltage and electrical resistance of the bipolar region during the electric current flow. In this case, the term “overvoltage” means the difference between the voltage on the bipolar membrane in a system under current and the voltage on the membrane in the absence of current. Simons [59,60] and Greben' [12,64] first showed that ionogenic groups of membranes can accelerate the reaction of water dissociation in the membrane system. Simons [59,60] proposed a mechanism of catalytic reaction. The rate constants of the limiting stages of the dissociation reaction of water molecules in ion exchangers contained in the space charge regions of bipolar membranes

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Nomenclature

E	integral specific energy consumption required to produce acid and alkali from sodium nitrate (kW h/kg)
$E_m(0)$	electric field strength at zero overvoltage of the bipolar region (V/m)
$E_m(U_b)$	electric field strength at the interface of cation and anion exchange layers at non-zero overvoltage of the bipolar region (V/m)
F	Faraday constant (96485 A s/mol)
I	current supplied to the electrodialyzer (A)
$-\mathrm{Im}Z$	imaginary part of the electrochemical impedance with the opposite sign (Ohm dm^2)
i	direct electric current density flowing through bipolar membrane (A/dm^2)
$i_{\sim}$	AC electric current density flowing through the studied membrane (A/dm^2)
$i_{H,OH}$	direct electric current density carried by hydrogen and hydroxyl ions generated by bipolar membrane (A/dm^2)
k_{Σ}	effective rate constant of catalytic reaction of water dissociation in the space charge region (in the bipolar region) at zero overvoltage of the bipolar region (s^{-1})
k_L	rate constant of the limiting stage of water dissociation reaction (s^{-1})
M	molar mass of nitric acid or sodium hydroxide (kg/mol)
$N_{el.cel}$	the number of unit cells in the membrane stack of the electrodialyzer
n	quantity of acid or alkali (mol)
P	the integral specific productivity of the apparatus for acid and alkali in terms of 1 m^2 of membrane active area (kg/(h m^2))
R	differential resistance of the studied membrane (Ohm dm^2)
R_b	differential resistance of the bipolar region (Ohm dm^2)
$\mathrm{Re}Z$	real part of electrochemical impedance (Ohm dm^2)
R_0	differential resistance obtained by extrapolation the low-frequency part of the impedance spectrum in which the dissociation of water molecules in the bipolar membrane is appeared at the ReZ axis (Ohm dm^2)
R_{∞}	differential resistance corresponding to the minimum of the imaginary part of the electrochemical impedance in the high-frequency part of the spectrum (Ohm dm^2)
S	active area of each membrane (m^2)
$U_{\sim}$	AC voltage applied to the studied membrane (V)
U_b	overvoltage of the bipolar region – the difference between potential drop under direct current and potential drop at zero current (V)
U_{EDS-1}	voltage supplied to the electrodialyzer (V)
$U_{el.cel}$	voltage supplied to the electrodialyzer containing electrode chambers separated by a cation exchange membrane at the same electric current at which U_{EDS-1} was measured (V)
Z	electrochemical impedance (Ohm dm^2)

Greek symbols

β	entropy factor (m/V)
ε	relative permittivity in the space charge region
ε_0	electric constant (8.854 F/m)
η	acid or alkali current efficiency
τ	time elapsed since the start of the main experiment (s)

increase in a range [63,65]:

$$-N(CH_3)_3 < -SO_3H < -PO_3H^- < =NH, -NH_2 < \equiv N < -COO^- < -PO_3^{2-}$$

k_L, s^{-1} 0 $3\cdot10^{-3}$ $3\cdot10^{-2}$ 10^{-1} 1 10 10^2

which is consistent with the results of determining the effective rate constants of the catalytic reaction in bipolar membranes with various ionogenic groups and catalytic additives [63,65].

The method for calculating the effective constants of a catalytic reaction using the method of electrochemical impedance was proposed in [66] and is based on the equation proposed by Timashev in [67–69]. The effective rate constants of the catalytic reaction with various ionogenic groups and catalytic inorganic additives in bipolar membranes were determined in [9,66].

The electromembrane processes of the recovery of salts into the corresponding acids and alkalis have been studied with homogeneous and heterogeneous membranes. The process of producing hydrochloric acid and sodium hydroxide in an electrodialyzer with bipolar membranes was studied in [2,12,13,70], sulfonic acid and sodium hydroxide from sodium sulfate in [71–73], nitric acid and sodium hydroxide from sodium nitrate in [73–75], nitric acid and ammonia from ammonium nitrate in [76], a mixture of nitric and hydrofluoric acids from neutralized effluents formed during steel pickling in [11,14,15].

Table 1
Types of initial cation-exchange and anion-exchange membranes used to produce of industrial and modified bipolar membranes [77]

Type of BPM	Layers of BPM [40,77]: Cation exchange layer Anion exchange layer Catalytic additive	Type of ion exchanger [77]	Ionogenic groups in the layer or additive [38,40]	BPM thickness, mm, not more than [38]	Total exchange capacity of the ion exchanger, mg-eq/ml [80,81]
MB-1	MC-40	KU-2	$-SO_3H$	1	1.8
	MA-40	EDE-10p	$\equiv N$, $=NH$, $-NH_2$		2.3
	—	—	—		—
MB-2	MC-40	KU-2	$-SO_3H$	0.9	1.8
	MA-41	AV-17	$-N^+(CH_3)_3$		1.0–1.2
	—	—	—		—
MB-3	MC-41	KF-1	$-PO_3H_2$	1	5 mg-eq/g [82]
	MA-41	AV-17	$-N^+(CH_3)_3$		1.0–1.2
	—	—	—		—
MB-2m	Ralex CMH	—	$-SO_3H$	0.66**	—
	Ralex AMH	—	$-N^+(CH_3)_3$		—
	Powder of ion-exchanger*	KF-1	$-PO_3H_2$		5 mg-eq/g [82]

* – Powder of ion-exchanger was used in the bipolar MB-2m membrane developed by the authors of this study
** – Thickness of MB-2m bipolar membrane was measured experimentally

In this work the electrochemical characteristics of the Russian industrial heterogeneous MB-1, MB-2, MB-3 bipolar membranes, and a modified MB-2m membrane containing an ion-polymer with catalytically active phosphoric acid groups [55], and the recovery of nitric acid and hydroxide sodium from a solution of sodium nitrate using these bipolar membranes were studied.

2. Materials and methods

2.1. Materials

Industrial heterogeneous MB-1, MB-2, and MB-3 bipolar membranes [77] and the modified heterogeneous MB-2m bipolar membrane (Table 1) containing the ion-polymer with catalytically active phosphoric acid groups in the bipolar region [55] were used as bipolar membranes in this work. Phosphoric acid groups provide the lowest voltage at the bipolar membrane due to the significant acceleration of the dissociation reaction of water molecules. Heterogeneous cation-exchange membranes MC-40 and anion-exchange membranes MA-41 were used as auxiliary ion-exchange membranes to measure the frequency spectra of electrochemical impedance [77]. Heterogeneous cation exchange membranes Ralex CMH and anion exchange membranes Ralex AMH [40] were used to study the process of producing nitric acid and sodium hydroxide from sodium nitrate.

In this study the modified bipolar MB-2m membrane was obtained by hot pressing of heterogeneous industrial Ralex CMH cation exchange membrane and Ralex AMH anion exchange membrane. The catalytic additive of ion-polymer containing phosphoric acid groups was applied to the surface of the cation exchange membrane before pressing to reduce the voltage at the bipolar membrane. The phosphoric acid cationite KF-1 obtained on the basis of a styrene copolymer with 8% divinylbenzene [43, Table 1] with a particle size of less than 30 μm was used as a catalytic additive. The KF-1 powder is applied to the surface of the cation exchange membrane as a wet paste to a thickness of 0.2 mm. In preliminary experiments it was found that with an increase in the mass of the KF-1 powder applied at masses less than 8 g/m^2 the voltage on the bipolar MB-2m membrane decreases, and with masses greater than 8 g/m^2 the voltage practically does not change. Therefore, the mass of KF-1 powder applied was 8 g/m^2. The concentration of phosphoric acid groups catalytically active in the dissociation reaction of water molecules in the bipolar region is determined by the total exchange capacity of the KF-1 cationite and equals 5 mg-eq/g of dry ionite.

The electron microscope images of the splits of bipolar membranes were obtained to determine the degree of sharpness of the transition from the cation exchange layer to the anion exchange layer. The location of sodium and chloride ions, which are counterions in the cation exchange and anion exchange layers, and phosphorus atoms, contained in the ionogenic groups of the catalytic additive, was determined by local X-ray spectral microanalysis (Fig. 1).

Fig. 1 shows that the boundary between the cation exchange grains (including grains of catalytic additive with phosphoric acid groups) and anion exchange grains where reaction of water molecules dissociation proceeds are defined in the bipolar region.

Green dots indicate chlorine, yellow dots indicate sodium, and red dots indicate phosphorus.

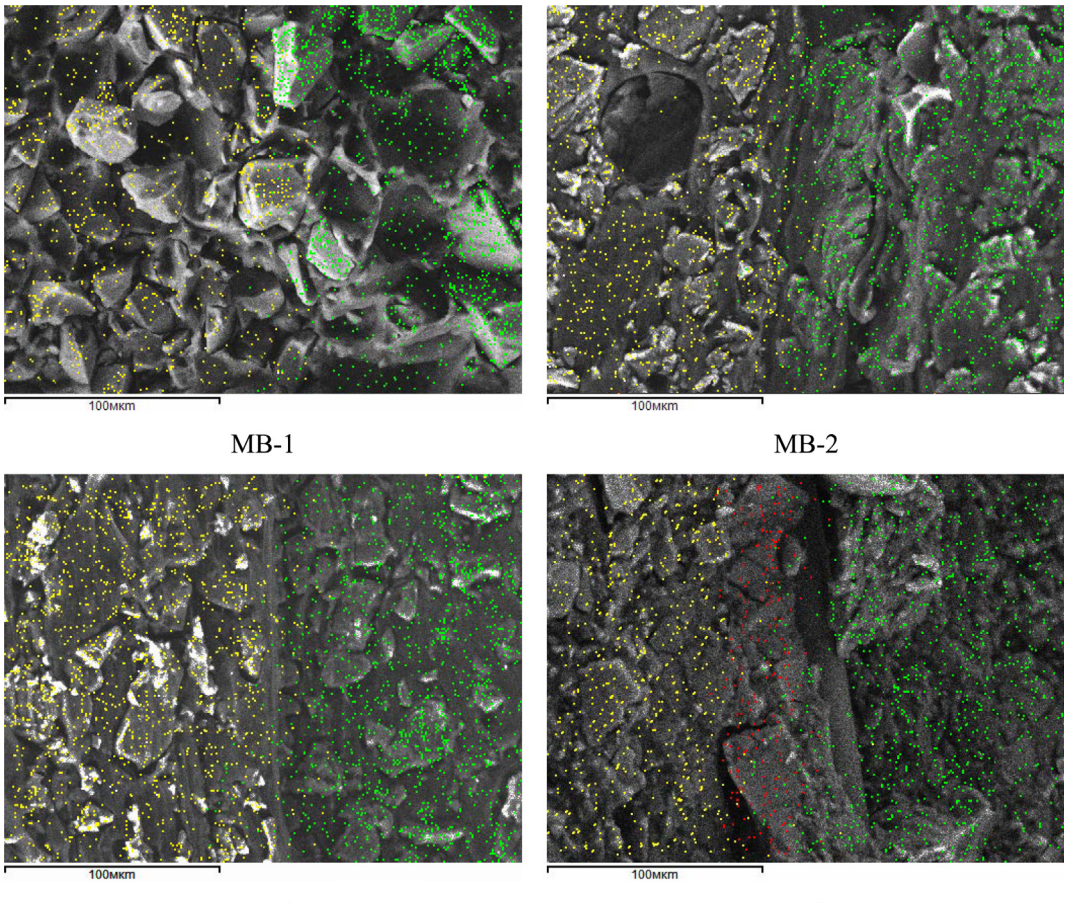


Fig. 1. Electron microscope images of the bipolar boundary region in the bipolar membrane splits. The cation exchange layer in sodium form is on the left side, anion exchange layer in chloride form is on the right side in each image. The particles of ion-polymer containing phosphoric acid groups, are located in the bipolar region of the bipolar membrane. The presence of chloride, sodium and phosphorus atoms was detected by local X-ray spectral microanalysis.

2.2. Methods of investigation of bipolar membranes

The electrochemical characteristics of the heterogeneous BPM (Table 1) were investigated in a system containing 0.1 M HNO_3 and 0.1 M NaOH by the electrochemical impedance technique in a four-electrode electrochemical cell (Fig. 2). The measurements were carried out in the stationary state both in the absence of a superimposed direct electric current and during its flow through the BPM in the mode of generation of hydrogen and hydroxyl ions. The frequency spectra of electrochemical impedance were measured in the frequency range of alternating current from 0.1 Hz to 1 MHz, evenly distributed on a logarithmic scale using an AUTOLAB 100 N potentiostat–galvanostat with an FRA32M impedance measurement module.

The typical frequency spectrum of the electrochemical impedance of a bipolar membrane has the form of a distorted semicircle (spectrum of the bipolar region), on the left side of which there is a rise associated with the geometric capacity of the membrane, shunted by the resistance of the monopolar layers of the bipolar membrane (Fig. 3).

The crossing of the low-frequency part of the spectrum with the abscissa corresponds to the resistance of the bipolar membrane at zero frequency R_0. The real part of the impedance at the minimum point of the spectrum in the left, high-frequency, part corresponds to the sum of the resistances of the monopolar regions R_∞. The resistance of the bipolar region R_b was found by the difference between R_0 and R_∞ [78]

$$R_b = R_0 - R_\infty \quad (1)$$

where R_0 is differential resistance, obtained by extrapolating the low-frequency part of the impedance spectrum, where the dissociation of water molecules in the bipolar membrane appears, at the ReZ axis; R_∞ is the differential resistance corresponding to the minimum of the imaginary part of the electrochemical impedance in the high-frequency part of the spectrum.

The differential resistance is measured by impedance spectroscopy and is described by the Eq. (2):

$$R = \frac{dU_\sim}{di_\sim} \quad (2)$$

where R – differential resistance of the studied membrane; $U_\sim$ – AC voltage applied to the studied membrane; $i_\sim$ – AC electric current density flowing through the studied membrane.

Eq. (3) for calculating the overvoltage of the bipolar region of the bipolar membrane by the known dependence of the resistance of the bipolar region on the current density arises from Eq. (2):

$$U_b = \int_0^{i^*} R_b \, di \quad (3)$$

The calculated overvoltages of the bipolar region were used to obtain a partial current-overvoltage characteristic of the bipolar region of the membrane. This characteristic confirms the efficiency of catalytically active ionogenic groups in the dissociation reaction of water molecules in the bipolar region of the membrane. The partial current-overvoltage characteristics of bipolar membrane regions were used to calculate the effective dissociation rate constants of water molecules k_Σ and the parameter β in the bipolar regions of the studied membranes by the Eq. (4) [66]

$$i_{H,OH} = k_\Sigma \frac{\varepsilon\varepsilon_0}{\beta}[\exp(\beta E_m(U_b)) - \exp(\beta E_m(0))] \quad (4)$$

where $i_{H,OH}$ is direct electric current density, carried by hydrogen and hydroxyl ions, generated by the bipolar membrane (in this work $i_{H,OH}$ is approximately assumed to be equal to direct electric current density i flowing through the bipolar membrane, due to the low concentration of acid and alkali solutions near the bipolar membrane (0.1 mol/L)); k_Σ is the effective rate constant of catalytic reaction of water dissociation in the space charge region (in the bipolar region) at zero overvoltage of the bipolar region; $E_m(U_b)$ is the electric field strength at the interface of cation- and anion exchange layers at nonzero overvoltage of bipolar region; U_b is the bipolar region overvoltage, calculated as the difference between potential drop under direct current and potential drop at zero current; $E_m(0)$ is the electric field strength at zero overvoltage of bipolar region; β is the entropy factor; ε is the relative permittivity in the space charge region; ε_0 is the dielectric constant.

The parameters k_Σ and β, included in Eq. (4), were found by approximating the partial current-overvoltage characteristic of the bipolar region [66] by minimizing the sum of squared residuals (a residual being: the difference between an observed value, and the fitted value provided by a model according to the Eq. (4)).

2.3. Bipolar membrane electrodialysis equipment and methods

The electrodialysis process of producing nitric acid and sodium hydroxide from sodium nitrate was studied in an electrodialyzer-synthesizer EDS-1 the membrane stack of which contained five three-chamber elementary cells (Fig. 4). Each unit cell of the EDS-1 included salt, alkaline and acid chambers separated by RALEX CMH cation exchange membrane, MB-2m bipolar membrane, and RALEX AMH anion exchange membrane.

The main processes of ion transfer across membranes in an electrodialyzer are shown by solid arrows and the undesirable processes of ion transfer across membranes are shown by dotted arrows.

The active area of each membrane and the electrodes was equal to 1 dm^2, the distance between the membranes was determined by the thickness of the frames and spacers and was 1 mm. The process was performed in a galvanostatic mode at a current density of 2 A/dm^2. A MATRIX MPS-3005L-1 DC power supply was used as the current source. The experiments were performed at an ambient temperature of 25 ± 2 °C. The batch-mode was used in the research BMED process, the linear velocity of the solutions through each chamber was 1 cm/s and was provided by a Heidolph PD 5001 peristaltic pump. The concentration of sodium nitrate in a salt solution was maintained in the range of 0.50 ± 0.05 mol/L.

The concentration of sodium nitrate and sulfonic acid in the salt chamber solution was determined by the measured values of the specific electric conductivity of the solution and its pH. For this purpose we used the calibration dependencies which were experimentally found at a temperature of 25 °C: a) dependence of the specific electric conductivity of the solution on the concentration of sodium nitrate and pH and b) dependence of the pH of the solution on the concentration of sodium nitrate and sulfonic acid. The concentration of sodium nitrate in

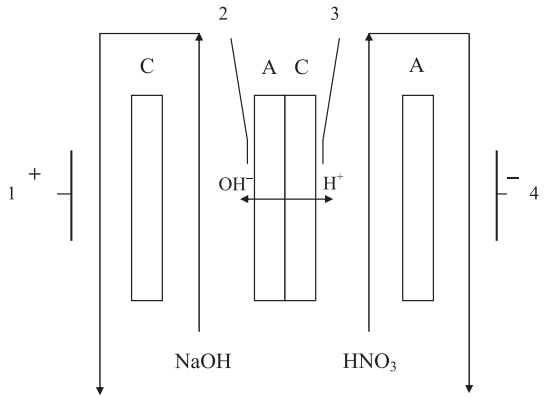


Fig. 2. Scheme of the electrochemical cell for measuring the frequency spectra of the electrochemical impedance of a bipolar membrane. C is cation exchange membrane MC-40; A is anion exchange membrane MA-41; AC is the bipolar membrane under investigation, 1 and 4 are electrodes polarizing the investigated bipolar membrane under investigation by direct and alternating current, 2 and 3 are potential electrodes for measuring the potential difference at the membrane by alternating current.

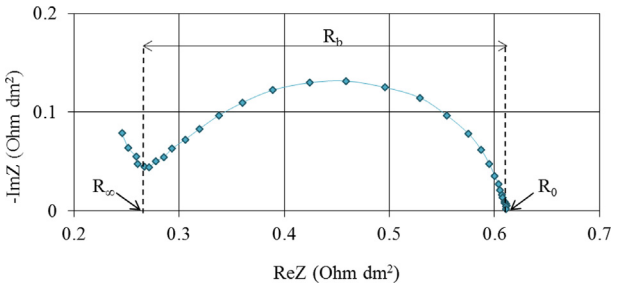


Fig. 3. Typical frequency spectra of electrochemical impedance of bipolar membrane. R_0 – differential resistance obtained by extrapolation the low-frequency part of the impedance spectrum in which the dissociation of water molecules in the bipolar membrane is appeared at the ReZ axis; R_∞ – differential resistance corresponding to the minimum of the imaginary part of the electrochemical impedance in the high-frequency part of the spectrum; R_b – differential resistance of the bipolar region.

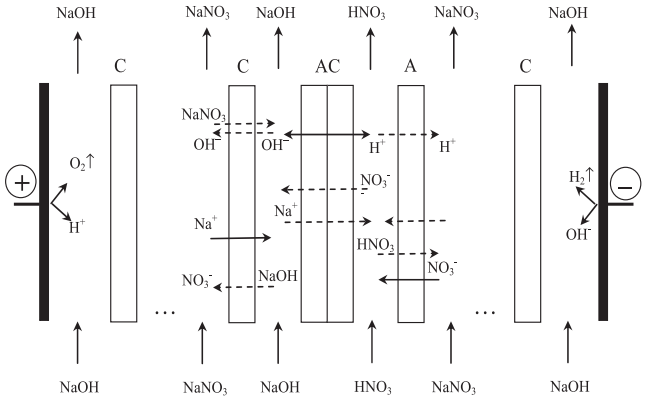


Fig. 4. Scheme of membrane stack of electrodialysis apparatus designed to produce nitric acid and sodium hydroxide from sodium nitrate, and the fluxes of ions and molecules, carried across the membrane by passing an electric current through the apparatus.

the salt solution was found from the first dependence and the concentration of sulfonic acid in the solution was found from the second dependence.

Initial concentrations of nitric acid and sodium hydroxide were 0.1 M. The initial volume of sodium nitrate solution was 4.2 L, the initial volumes of nitric acid and sodium hydroxide solutions were equal to 1.2 L. The volumes of solutions were measured during the experiment using graduated reservoirs. The volumes of solutions in the connecting tubes were also taken into account in the calculations. The temperature of salt, acid and alkali solutions was measured with thermometers and increased during the experiment from 25 ± 2 °C to 29 ± 2 °C, 30 ± 2 °C and 30 ± 2 °C respectively.

The concentrations of sodium hydroxide and nitric acid in solution samples were determined by a Titroline 6000 automatic titrator. The quantities of acid and alkali were calculated from the found concentrations and measured volumes of acid and alkali solutions, circulating through the EDS-1. The basic electrochemical characteristics of the apparatus were determined according to the dependence of the quantities of acid and alkali on time.

The integral current efficiencies of acid and alkali in the electrodialysis apparatus were calculated by the Eq. (5)

$$\eta = \frac{F}{IN_{\text{el.cel}}}\left(\frac{n}{\tau}\right) \tag{5}$$

where I is the current supplied to the apparatus; $N_{\text{el.cel}}$ is the number of elementary cells in the membrane stack of the electrodialysis apparatus; n is the quantity of the produced acid or alkali; τ is the time passed since the start of the main experiment; F is Faraday constant.

The integral specific energy consumptions required to produce acid or alkali E were calculated by the Eq. (6)

$$E = \frac{(U_{\text{EDS}-1} - U_{\text{el.cel}})F}{M\eta} \tag{6}$$

where $U_{\text{EDS-1}}$ is the voltage supplied to the electrodialysis apparatus; $U_{\text{el.cel}}$ is the voltage, supplied to the electrodialysis apparatus containing electrode chambers separated by a cation exchange membrane at the same electric current at which $U_{\text{EDS-1}}$ is measured; M is the molar mass of nitric acid or sodium hydroxide.

The integral specific productivities of the apparatus for acid or alkali were calculated by the Eq. (7):

$$P = \frac{\eta IM}{FS} \tag{7}$$

where S is active area of each membrane.

The concentrations of impurities (nitrate anions in alkali and sodium ions in nitric acid) were determined by the Aquilon Stayer ion chromatograph.

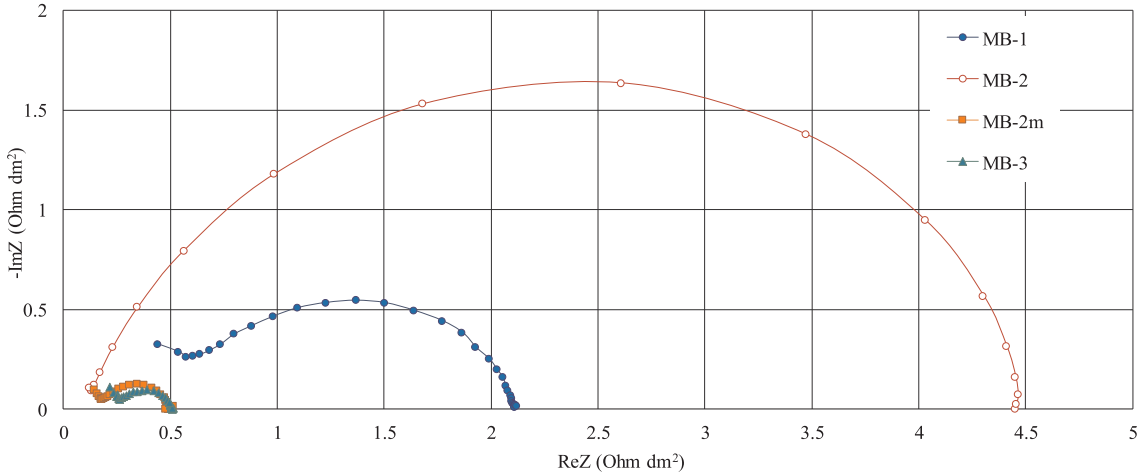


Fig. 5. Frequency spectra of electrochemical impedance of MB-1, MB-2, MB-2m, and MB-3 bipolar membranes in 0.1 M HNO_3 | 0.1 M NaOH system at 0.88 A/dm^2 direct current density.

3. Results and discussion

3.1. Electrochemical characteristics of heterogeneous bipolar membranes

It is evident from the obtained electrochemical impedance spectra of industrial heterogeneous MB-1, MB-2, MB-3 membranes, and the modified heterogeneous MB-2m membrane, that the resistance of the bipolar region ($R_b = R_0 - R_\infty$) of the modified membrane is an order of magnitude lower than the resistance of the bipolar region of the original MB-2 (Fig. 5).

Industrial MB-2 and modified MB-2m bipolar membranes contain strong-acid sulfonic acid groups in the cation exchange layers and strong-base quaternary ammonium groups in anion exchange layers (Table 1). The decrease of R_b at MB-2m membrane is explained by the presence in its bipolar region of particles of an ion-polymer containing phosphoric acid groups which have high catalytic activity in the reaction of dissociation of water molecules [9].

It is seen (Fig. 6) that the shape of the frequency spectra of the bipolar membrane MB-2m at all current densities in the studied frequency range is the same, and the resistance of the bipolar region of the membrane MB-2m depends on the direct current density. An increase in the resistance of the bipolar region at low (0–0.04 A/dm^2) currents is explained by a decrease in the concentration of mobile ions and an increase in the thickness of the space charge at the cation/anion interface [9,79] at increasing current. At higher than 0.04 A/dm^2 currents this resistance decreases due to an increase in the concentration of H^+ and OH^- ions in the space charge region. The presence of a maximum on the dependence of the resistance of the bipolar region on the current density is a common feature for industrial MB-1, MB-2, and MB-3 bipolar membranes too.

The resistance of the bipolar region and the partial current-overvoltage characteristic of the MB-2m membrane are close to the corresponding characteristics of the MB-3 membrane (Fig. 7) and significantly differ from the characteristics of the MB-1 and MB-2membranes.

The effective dissociation rate constants of the water in the space charge region (in the bipolar region) k_Σ for the studied membranes (Table 2) show that the dissociation of water molecules in industrial MB-1, MB-2, and MB-3 membranes runs many times slower than in the MB-2m bipolar membrane. This indicates a high activity of the catalytic additive in the dissociation reaction of water molecules in the space charge region of the bipolar MB-2m membrane (Table 2). The values of the dissociation rate constants in the studied membranes are consistent with the series proposed in [63,65].

To estimate the concentration range for nitric acid and sodium hydroxide and the range of current densities in which the MB-2m bipolar membrane can be used in the processes of producing nitric acid and sodium hydroxide from sodium nitrate, the dependences of the resistance on current density and the current-overvoltage characteristics of the bipolar region on the concentrations of adjacent solutions of nitric acid and sodium hydroxide were calculated. As the concentration of acid and alkali solutions increases, the resistance of the bipolar region and the overvoltage on it decrease due to an increase in the concentration of salt ions in the bipolar region (Fig. 8). This effect is undesirable, as it is explained by the transport of salt ions and reduces the current efficiency of hydrogen and hydroxyl ions. Undesired transport of salt ions is strongly pronounced in the region of low current densities and practically disappears at high current densities exceeding 2 A/dm^2. Thus, the MB-2m bipolar membrane can be used above 2 A/dm^2 current densities if the concentration of nitric acid and sodium hydroxide is lower than 1 mol/L.

3.2. Electrochemical characteristics of the electromembrane process for the recovery of nitric acid and sodium hydroxide from sodium nitrate solution

A comparison of the dependences of the concentrations of sodium hydroxide and nitric acid on the operating time shows that the concentrations of solutions produced with industrial MB-1 and MB-3 bipolar membranes are lower than produced in the EDS with the modified MB-2m bipolar membrane (Fig. 9). So, at times of 300 min or later the increase in the concentration of acid and alkali slows down significantly in the case of MB-1, MB-3 membranes and continues to grow in the case of a modified MB-2m membrane approaching the concentrations of acid and alkali solutions produced in the apparatus with homogeneous BMP1 membranes [78, Fig. 2]. In addition, the resulting concentrations of nitric acid are always lower than alkalis due to the greater non-selective transport of hydrogen ions into salt chambers from acid chambers through an anion exchange membrane than non-selective transport of hydroxyl ions from alkaline chambers through a cation exchange membrane (Fig. 4).

The integral current efficiencies of acid and alkali also show that the process efficiency is higher in the case of EDS with the modified MB-2m bipolar membrane than in the case of MB-1 and MB-3 membranes (Fig. 10). However, MB-2m bipolar membrane has overvoltage of the bipolar region a little greater than the MB-3 membrane (Fig. 7 b). Therefore, the current efficiency of hydrogen and hydroxyl ions on MB-2m membranes is much higher than on MB-3 membranes. In this regard, other characteristics of the electrodialyzer with MB-2m membranes (contamination of acid and alkali with the initial salt, specific energy consumption for acid and alkali production, specific acid and alkali performance) are much better than similar characteristics of the electrodialyzer with MB-3 membranes (Figs. 11–13). The current

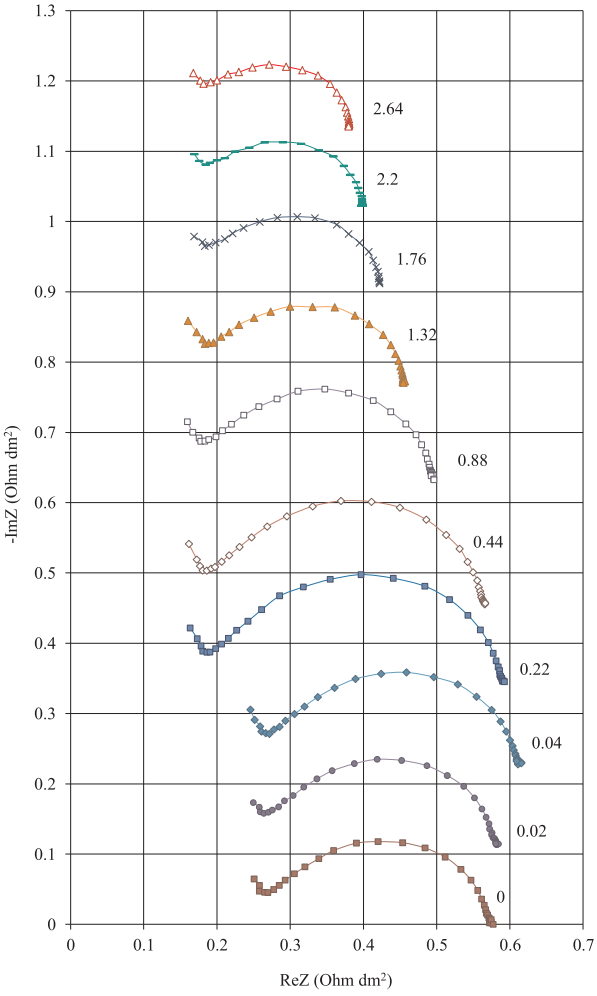


Fig. 6. Frequency spectra of the electrochemical impedance of MB-2m bipolar membrane in 0.1 M HNO_3|0.1 M NaOH system at various densities of direct electric current, A/dm^2.

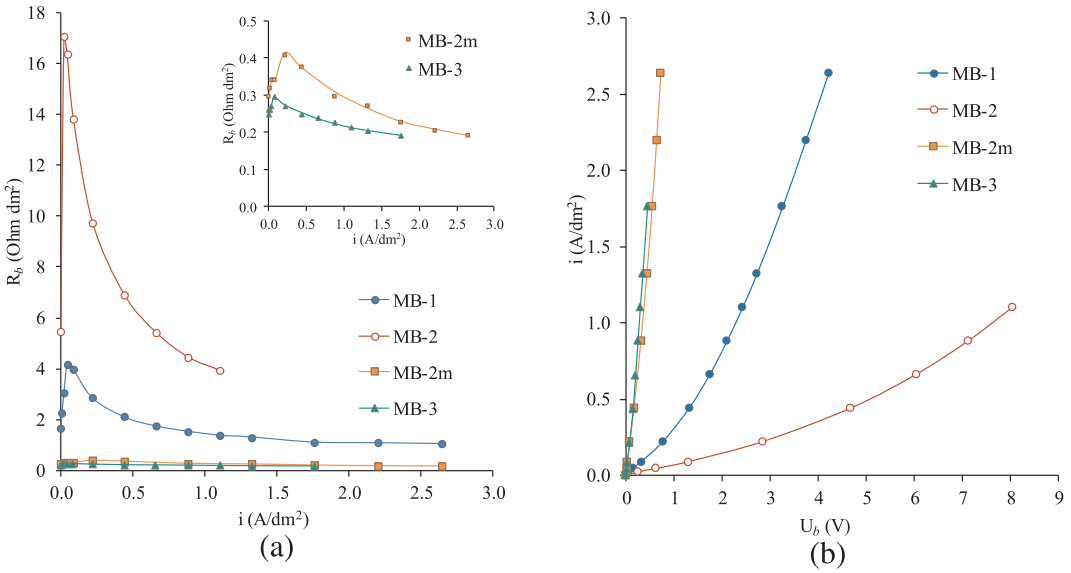


Fig. 7. The dependence of the bipolar region resistance of the membranes on the current density (*a*), and the partial current-overvoltage characteristics of the bipolar region of the membranes (*b*) in 0.1 M HNO_3|0.1 M NaOH system.

Table 2
Effective water dissociation rate constants k_Σ in the space charge region and the parameters β in the space charge region in the bipolar heterogeneous membranes in 0.1 M HNO_3 | 0.1 M NaOH system calculated according to [66].

Type of BPM	k_Σ,1/s	β.10^9,m/V
MB-1	40.4 ± 2.3	3.22 ± 0.07
MB-2	8.7 ± 0.3	2.63 ± 0.03
MB-3	58 ± 16	7.3 ± 0.5
MB-2m	96 ± 9	5.2 ± 0.2

efficiencies of acid and alkali decrease with increases in their concentration which is associated with the diffusion of acid and alkali into the bipolar membrane, with the migration of hydrogen and hydroxyl ions as well as the diffusion of acid and alkali into the salt chamber from the acid and alkaline chambers.

The contamination of the produced nitric acid solution with sodium ions and sodium hydroxide solution with nitrate ions arises when the concentration of produced acid and alkali increases (Fig. 11). Increase of contamination is caused by the diffusion of acid and alkali into the bipolar membrane and the further transport of sodium ions through it into the acid solution and nitrate ions into the alkali solution as well as diffusion of sodium nitrate from salt solution into alkali solution and acid solution.

The contamination of sodium hydroxide with nitrate ions (Fig. 11 b) and nitric acid with sodium ions (Fig. 11 a) at concentrations above 0.5 mol/L is higher in the case of industrial MB-1, MB-3 membranes and lower when using the MB-2m bipolar membrane. The reason for these differences is the presence of strongly acidic sulfonic groups in the cation exchange layer and strongly basic quaternary ammonium bases in the anion exchange layer of the MB-2m bipolar membrane. As a result, the dissociation of ionogenic groups in the MB-2m membrane is less reduced in the presence of strong acid and alkali near the bipolar membrane than the dissociation of phosphoric acid groups in the MB-3 membrane and the dissociation of secondary and tertiary amino groups in the MB-1 membrane [12,79].

A decrease in the degree of dissociation of phosphoric acid groups in the cation exchange layer of the MB-3 membrane, which is in contact with a nitric acid solution, leads to decreases in the concentration of charged forms of phosphoric acid groups and the exchange capacity of

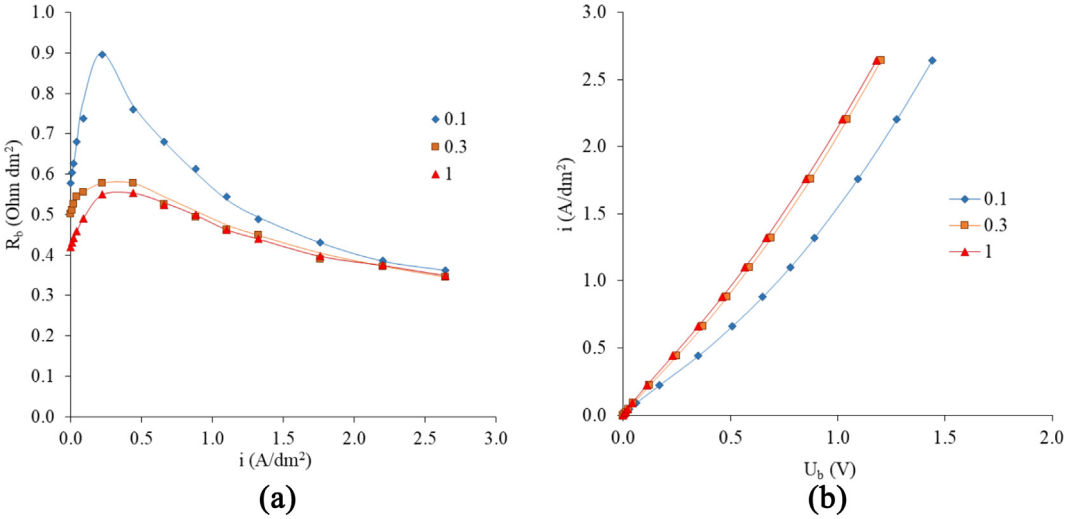


Fig. 8. The dependence of the bipolar region resistance on the current density (*a*) and the partial current–voltage characteristics (*b*) of the bipolar region MB-2m membrane at concentrations of nitric acid and sodium hydroxide equal to 0.1; 0.3 and 1 mol/L.

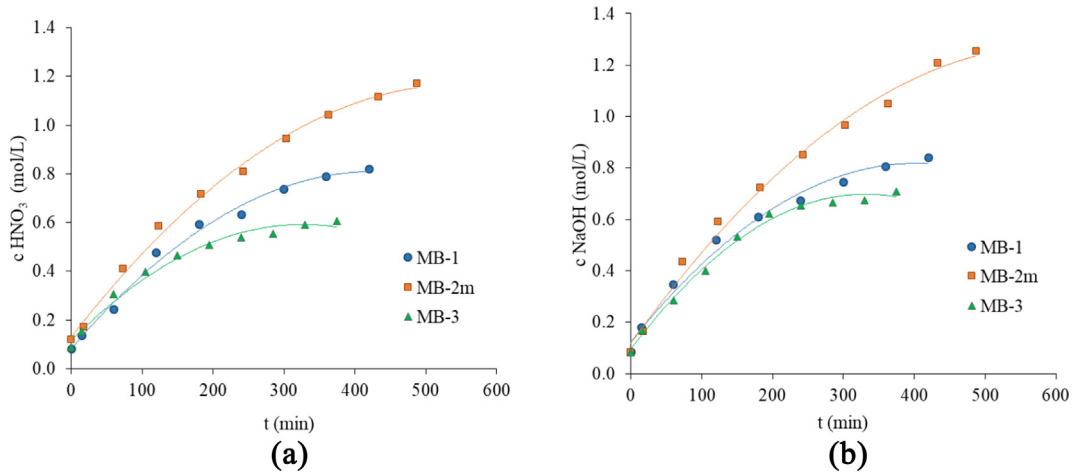


Fig. 9. Dependences of the concentrations of produced nitric acid (*a*) and sodium hydroxide (*b*) in solutions on the operating time of the EDS.

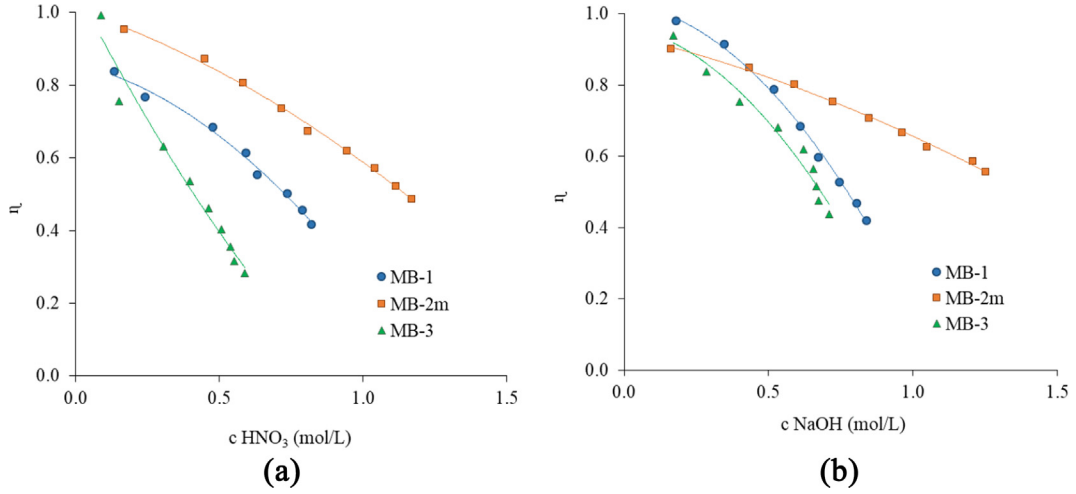


Fig. 10. Dependences of the integral current efficiencies of acid (*a*) and alkali (*b*) on the concentration of nitric acid and sodium hydroxide.

the cation exchange layer, and leads to a higher concentration of nitrate ions in the layer, which are coions, due to Donnan's sorption. An increase in the concentration of nitrate ions in the cation exchange layer leads to an increase in their transport number. Since the limiting stage of the nitrate ion transfer through the bipolar membrane is their transfer as coions through the cation exchange layer, their transfer number also increases through the whole bipolar membrane which in turn leads to contamination of the alkali obtained with nitrate ions (Fig. 10b). Similar phenomena are caused by an increase in the transport number of sodium ions across the whole bipolar membrane, which

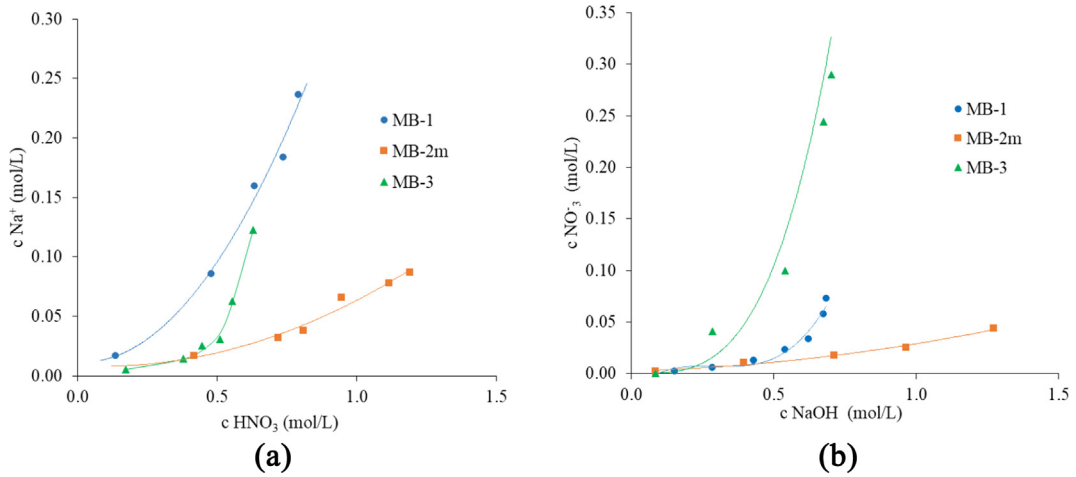


Fig. 11. Dependences of the concentration of sodium ions in nitric acid solution (*a*) and nitrate ions in sodium hydroxide solution (*b*) on the concentration of acid and alkali solutions produced in EDS containing MB-1, MB-3 or MB-2m bipolar membranes.

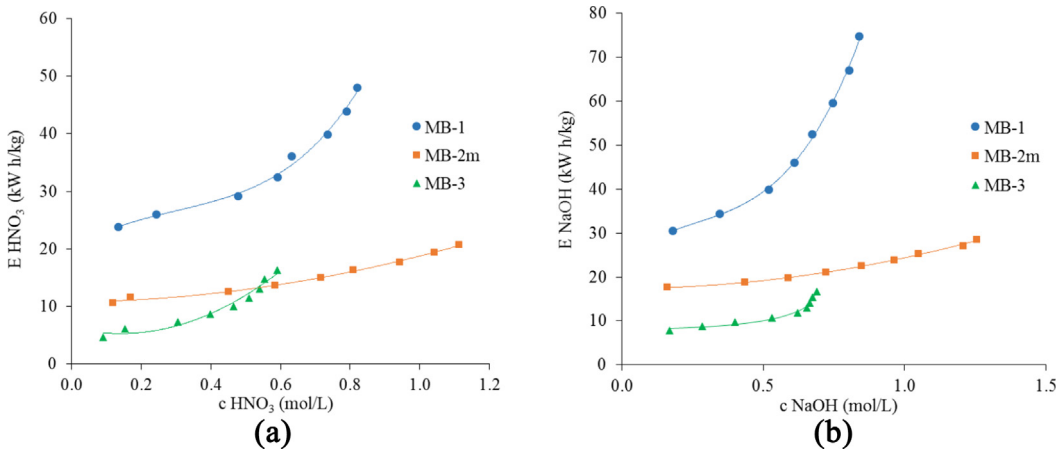


Fig. 12. Dependence of the integral specific energy consumption during production of nitric acid (*a*) and sodium hydroxide (*b*) from sodium nitrate on concentration of acid and alkali solutions.

in turn leads to contamination of the acid obtained with sodium ions (Fig. 10a). However, in this case, the dissociation of secondary and tertiary ionogenic groups in the anion-exchange layer of the MB-1 bipolar membrane, which is in contact with a sodium hydroxide solution, is suppressed.

An increase in the transport numbers of nitrate ions and sodium ions through the bipolar membrane leads to an increase in the fraction of the current spent on their transfer through the bipolar membrane, and, as a result, to a decrease in the fraction of the current spent on the transfer of hydrogen ions through the cation exchange layer of the bipolar membrane and hydroxyl ions through its anion-exchange layer, that is, to a decrease in their current efficiency on the bipolar membrane.

The integral specific energy consumption (Fig. 12) during alkali and acid production in the EDS with MB-1, MB-3 or MB-2m bipolar membranes shows that the modified MB-2m bipolar membrane provides the lowest energy consumption for producing acid and alkali at concentrations higher than 0.7–0.8 mol/L. At concentrations below 0.8 mol/L the electrodialyzer with an MB-3 bipolar membrane has lower energy consumption when producing alkali. However, with a further increase in the concentration of alkali obtained there should be a sharp increase in energy consumption when it is obtained (Fig. 11b) in the same way as in the case of acid (Fig. 11a). This fragment was not obtained in the experiment since at current concentrations of 0.7–0.8 mol/L the current efficiency in acid and alkali became low (less than 0.3 and about 0.4, respectively) and the experiment was terminated.

The integral specific productivity of the apparatus for nitric acid (Fig. 13a) as well as the integral current efficiency of acid (Fig. 10) is higher when using the MB-2m membrane than when using the MB-1 or MB-3 membranes in the whole range of concentrations. The reasons for this are the same as the reasons for the differences between these bipolar membranes in the current efficiency and specific energy consumption for acid and alkali producing.

4. Conclusions

The MB-2m bipolar membrane obtained by hot pressing from a Ralex CMH cation exchange membrane and Ralex AMH anion exchange membrane with the addition of a KF-1 cation exchanger powder containing phosphoric acid groups catalytically active in the water dissociation reaction has overvoltage of the bipolar region comparable with overvoltage of the bipolar region of the bipolar membrane MB-3.

The use of a Ralex CMH cation exchange membrane containing sulfonic acid groups and a Ralex AMH anion exchange membrane containing quaternary ammonium groups in the preparation of the MB-2m bipolar membrane makes it possible to obtain solutions of nitric acid and sodium hydroxide with a higher concentration than when using an MB-1 or MB-3 bipolar membrane. Other characteristics of an electrodialyzer with MB-2m membranes (current efficiency of acid and alkali, contamination of acid and alkali with initial salt, specific energy consumption for acid and alkali production, specific acid and alkali productivity) are also better than similar characteristics of electrodialyzers with MB-1 or MB-3 membranes. These differences are explained by a decrease in the degree of dissociation of phosphoric acid

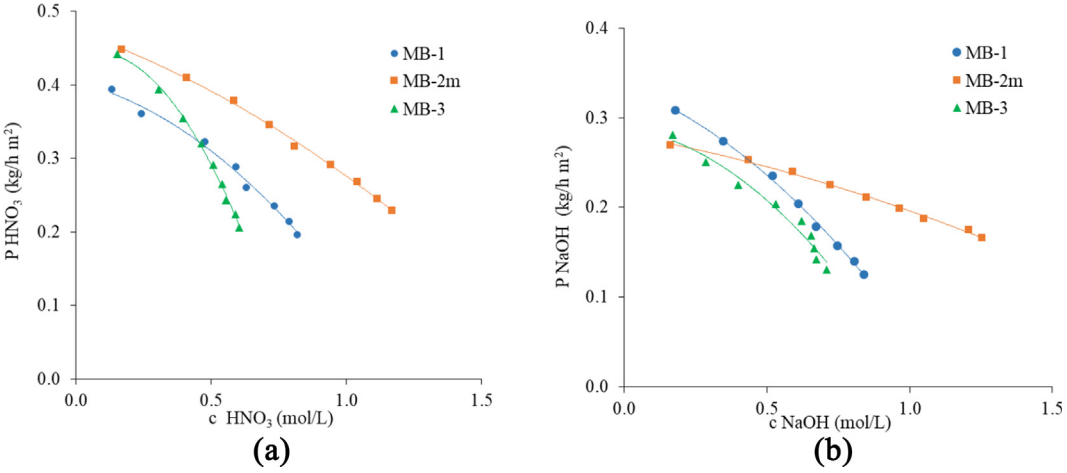


Fig. 13. Dependence of the integral specific productivity for nitric acid (*a*) and sodium hydroxide (*b*) on concentration of the produced acid and alkali solutions.

groups in the cation exchange layer of the MB-3 bipolar membrane in contact with the nitric acid solution and the depression of dissociation of secondary and tertiary amino groups in the anion exchange layer of the MB-1 bipolar membrane in contact with the sodium hydroxide solution. A decrease in the degree of dissociation of fixed ionogenic groups leads to a decrease of selectivity of the layers of the MB-1 and MB-3 bipolar membranes compared with the layers of the MB-2m membrane. As a result, the transport of nitrate and sodium ions through MB-1 and MB-3 bipolar membranes increases, the current efficiency of hydrogen and hydroxyl ions on the bipolar membranes decreases, and their other electrochemical characteristics deteriorate.

CRediT authorship contribution statement

N.V. Sheldeshov: Conceptualization, Methodology, Validation. **V.I. Zabolotsky:** Validation, Supervision. **N.V. Kovalev:** Investigation, Writing - review & editing, Visualization, Writing - original draft. **T.V. Karpenko:** Investigation, Writing - review & editing.

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